Little Lab Coats

🔍 What Is a Problem?

Engineering Design — Build It! | Unit 1, Lesson 1 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔧 **Subject:** Engineering Design — Build It! 🧰 **Kit Lesson:** No

Every invention starts with a problem. Engineers are professional problem-finders AND problem-solvers!

📋 **NGSS Standard: K-2-ETS1-1** — Ask questions, make observations, and gather information about a situation people want to change in order to define a simple problem that can be solved through the development of a new or improved object or tool.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Identify* a real problem from a described situation
- *Ask* clarifying questions about what makes a problem difficult
- *Explain* what a good solution needs to do (the criteria)

🔬 The Big Idea

Before engineers can build anything, they have to understand the *problem* — really, truly understand it. What's going wrong? Who does it affect? What would make it better? An engineer who doesn't fully understand the problem is just guessing. Good engineers ask *lots* of questions first!

Today, students will play the **"Spot the Problem" game** — hearing about real-life situations, identifying the problem, and deciding what a good solution would need to do.

> 💡 **Key vocabulary:** problem, solution, criteria (what a solution needs to do), engineer, design

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❓ Ask Questions, Make Observations

Engineering Design — Build It! | Unit 1, Lesson 2 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔧 **Subject:** Engineering Design — Build It! 🧰 **Kit Lesson:** No

Real engineers ask questions BEFORE they build. The more you understand the problem, the better your solution will be!

📋 **NGSS Standard: K-2-ETS1-1** — Ask questions, make observations, and gather information about a situation people want to change in order to define a simple problem that can be solved through the development of a new or improved object or tool.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Observe* a challenge situation closely before jumping to solutions
- *Ask* at least 3 good questions about the challenge
- *Gather* information by testing before designing

🔬 The Big Idea

Here's a secret that separates great engineers from average ones: **great engineers ask questions *first*.** Before touching a single piece of material, they want to understand everything about the problem. How big? How heavy? How fast? How many? What are the limits?

Today, we'll practice the most important [illegible]

> 💡 **Key vocabulary:** [illegible] *(how much tape you have)*, criteria, test

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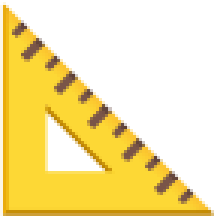
✏️ Draw Your Idea

Engineering Design — Build It! | Unit 1, Lesson 3 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔧 **Subject:** Engineering Design — Build It! 🧰 **Kit Lesson:** No



Real engineers draw their ideas before they build. A good sketch saves time, materials, and mistakes!

📋 **NGSS Standard: K-2-ETS1-2** — Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Create* a labeled sketch of a solution design
- *Label* each part of their design and explain what it does
- *Explain* their design to a parent/teacher before building

🔬 The Big Idea

Real engineers don't just start building and hope for the best. They dr eir design first — this is called a **blueprint** or **engineering drawing**. By drawing first, engineers can:

- See if their idea makes sense be
- Explain their idea to others and get feedback
- Plan exactly what th
- Notice problems be

🤯 **Real Engineering Facts:** be drawing and planning. The design

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💡 **Key vocabulary:** sketch, blueprint, label, design, materials, structure, function

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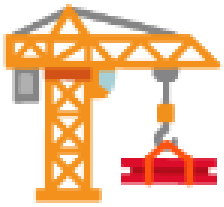
🏗️ Build It!

Engineering Design — Build It! | Unit 1, Lesson 4 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔧 **Subject:** Engineering Design — Build It! 🧰 **Kit Lesson:** No



Time to build! Remember your drawing from Lesson 3 — let's see how your plan comes to life!

📋 **NGSS Standard: K-2-ETS1-2** — Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Build* the bridge they designed in Lesson 3
- *Compare* the finished bridge to their original drawing
- *Describe* what was different, what surprised them, and what worked unexpectedly

🔬 The Big Idea

Now comes the exciting part — turning your drawing into a real, physical thing! But here's a secret that every engineer knows: **your first built version almost never looks exactly like your drawing.** That's completely normal! That gap between "what I drew" and "what I built" is actually super valuable — it teaches you things you couldn't have learned any other way.

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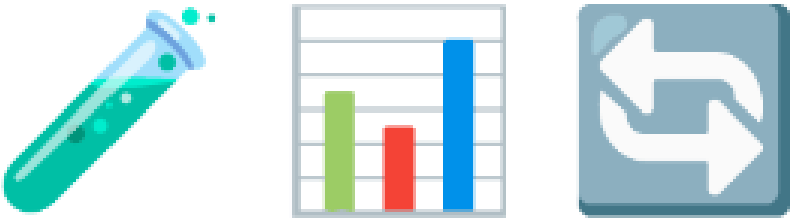
🧪 Test and Improve

Engineering Design — Build It! | Unit 1, Lesson 5 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔧 **Subject:** Engineering Design — Build It! 🧰 **Kit Lesson:** No



Testing tells you the truth. Every result — pass or fail — is information. And information makes your next design better!

📋 **NGSS Standard: K-2-ETS1-3** — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Test* their bridge against specific, measurable criteria
- *Record* test results accurately in a data table
- *Redesign* based on what the data shows

🔬 The Big Idea

In Lesson 4, you built your bridge and took a quick look at how it wo Today, we get serious about **testing**. Real engineers define exact criteria — the rules a design must pass — and then test their design ag st those rules. Results are *recorded*, not just remembered. Then the magic happens: we use the data to make a better design!

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🎤 Share Your Solution

Engineering Design — Build It! | Unit 1, Lesson 6 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔧 **Subject:** Engineering Design — Build It! 🧰 **Kit Lesson:** No

Real engineers present their work! Today you'll share your bridge, your process, and what you learned — just like a real engineering showcase!

📋 **NGSS Standard: K-2-ETS1-3** — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Present* their final bridge design to a family member
- *Explain* their complete engineering process from problem to solution
- *Evaluate* which version of their design worked best and *why*

🔬 The Big Idea

Engineering doesn't end when you build something. Real engineers communicate their work — they explain their designs, show their data, and teach others what they learned. This is called an **engineering presentation** or **design showcase**.

Today, students will present their bridge to a family member. This is their chance to show off everything they've done in this unit.

⭐ Welcome to the Engineering Showcase! ⭐

You designed, tested, and improved your bridge. Now it's your turn to share!

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💡 **Key vocabulary:** present, evaluate, compare, engineering report, documentation, communicate

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